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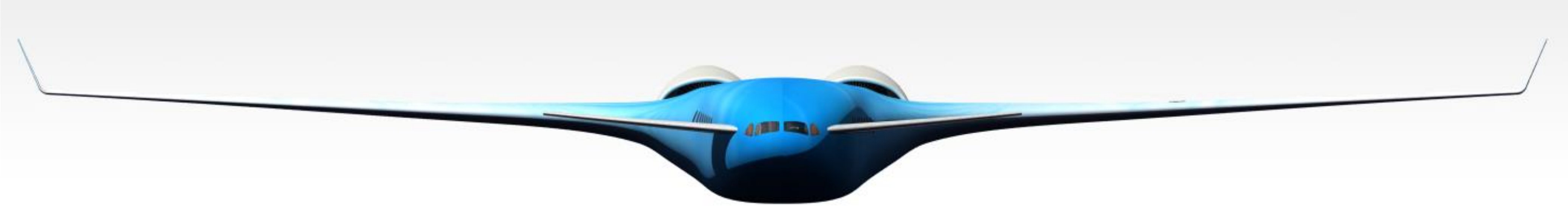

AE4238: Aero Engine Technology

Engine Inlets-I: The main functions

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TU Delft

10/21/2022

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What will we learn in this lecture

- What are Aircraft Inlets
- Some of design criteria
- Examples

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Engine Intake

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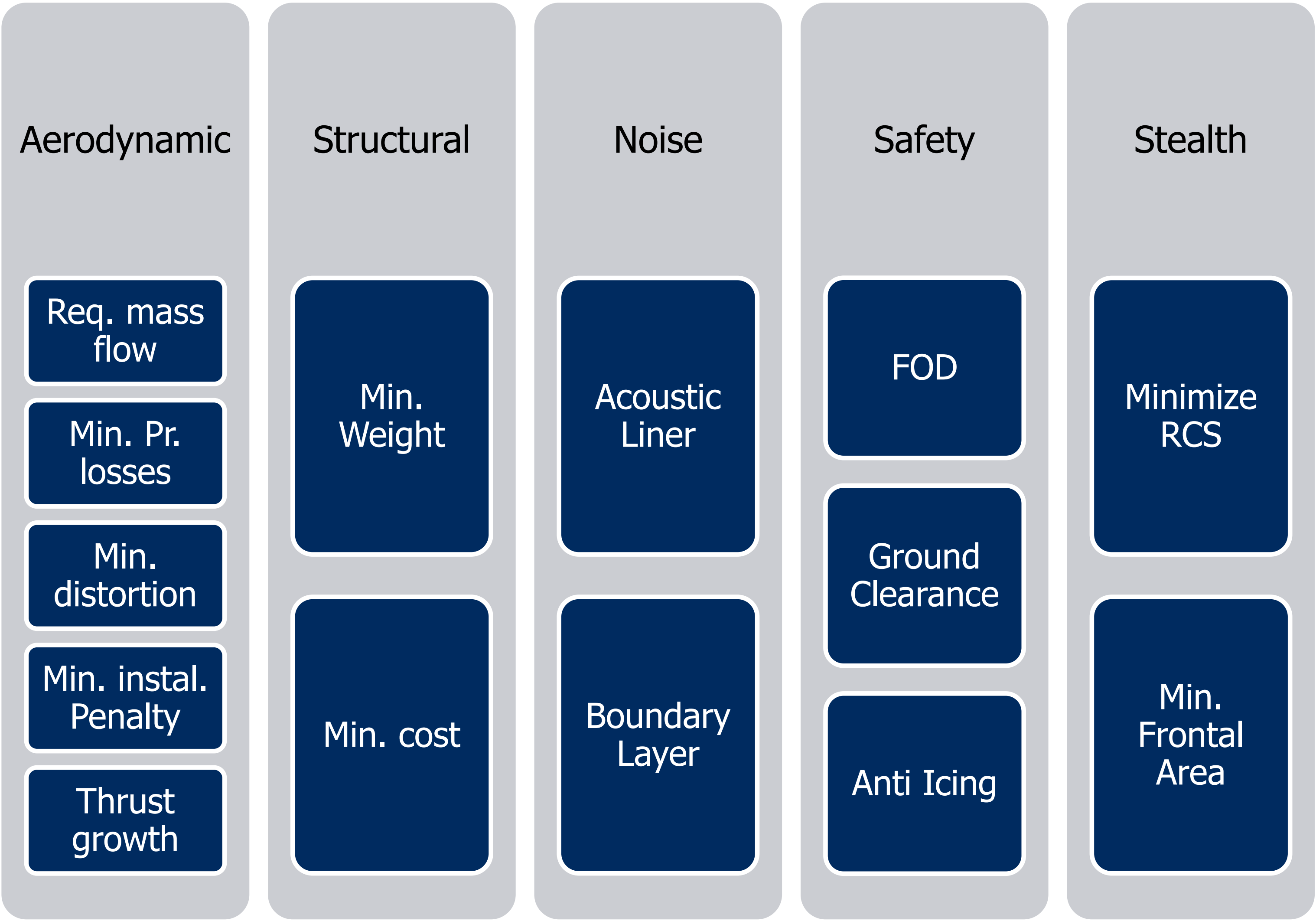
Intake design considerations

Intake design considerations



Foreign object damage

Jet engine intake

- Essentially a fluid flow duct whose task is to process airflow in a way that ensures the engine functions properly
- Provide adequate amount of uniform flow
- Cruising: conversion of E_{kin} into E_{pot} (pressure)
- Ram compression / “Ram recovery”
- Supersonic vs. subsonic intake
- Subsonic inlets are dominated by the boundary layer behavior.
- Supersonic inlets are dominated by the shock structure
- From 1D compressible aerodynamics
- $dA/A = (M^2-1)*(dV/V)$
- “Bellmouth” intake (test bed/stationary)

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Louvers Engine Intake (Mig 29)

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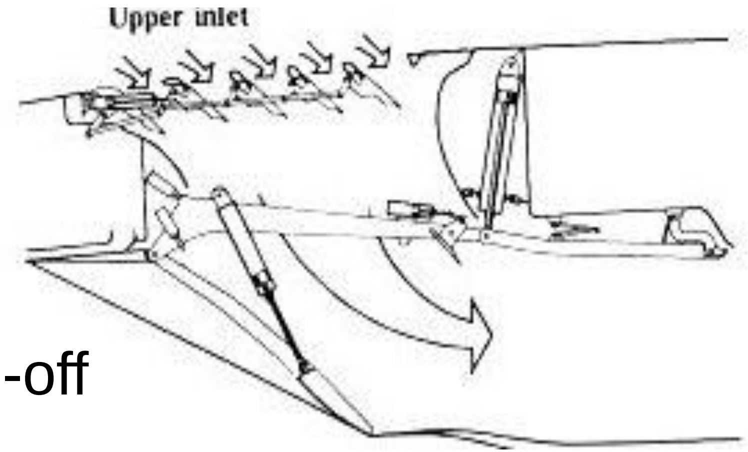
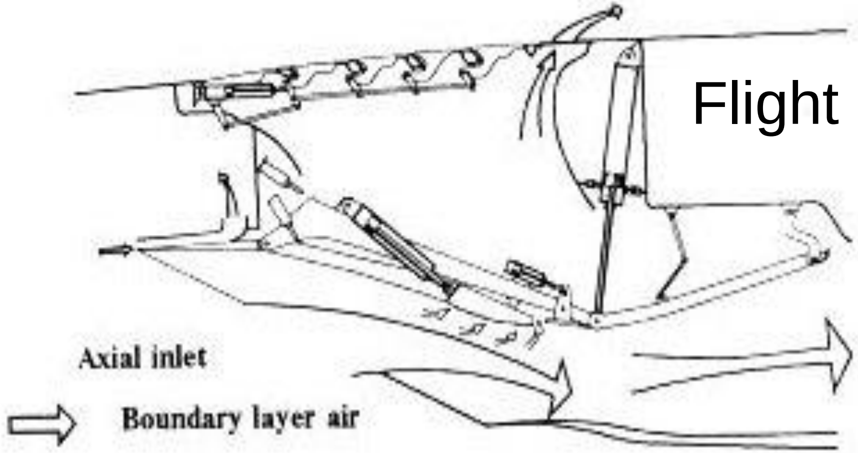
Mig 29 FOD protection

Inlet door (down) with porous surfaces for boundary layer bleed

Auxiliary inlet doors used during take-off



Take-off



Flight

Over the wing Intake (F117A)

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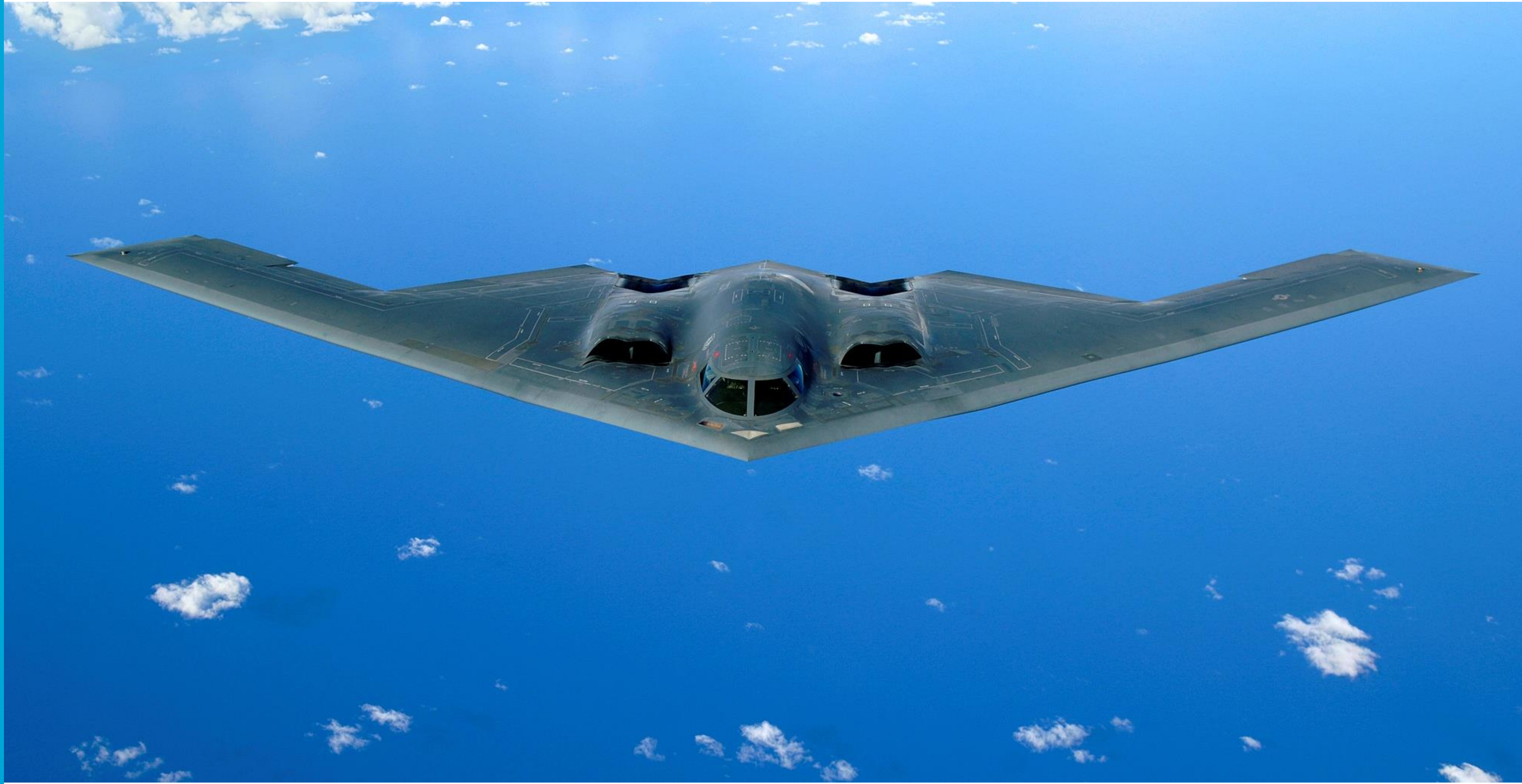
Shielded Engine Intake (B2)



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Minimum Ground Clearance

Boeing 737-800 NG

Boeing 737-800 Max

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Subsonic inlet (GE90)

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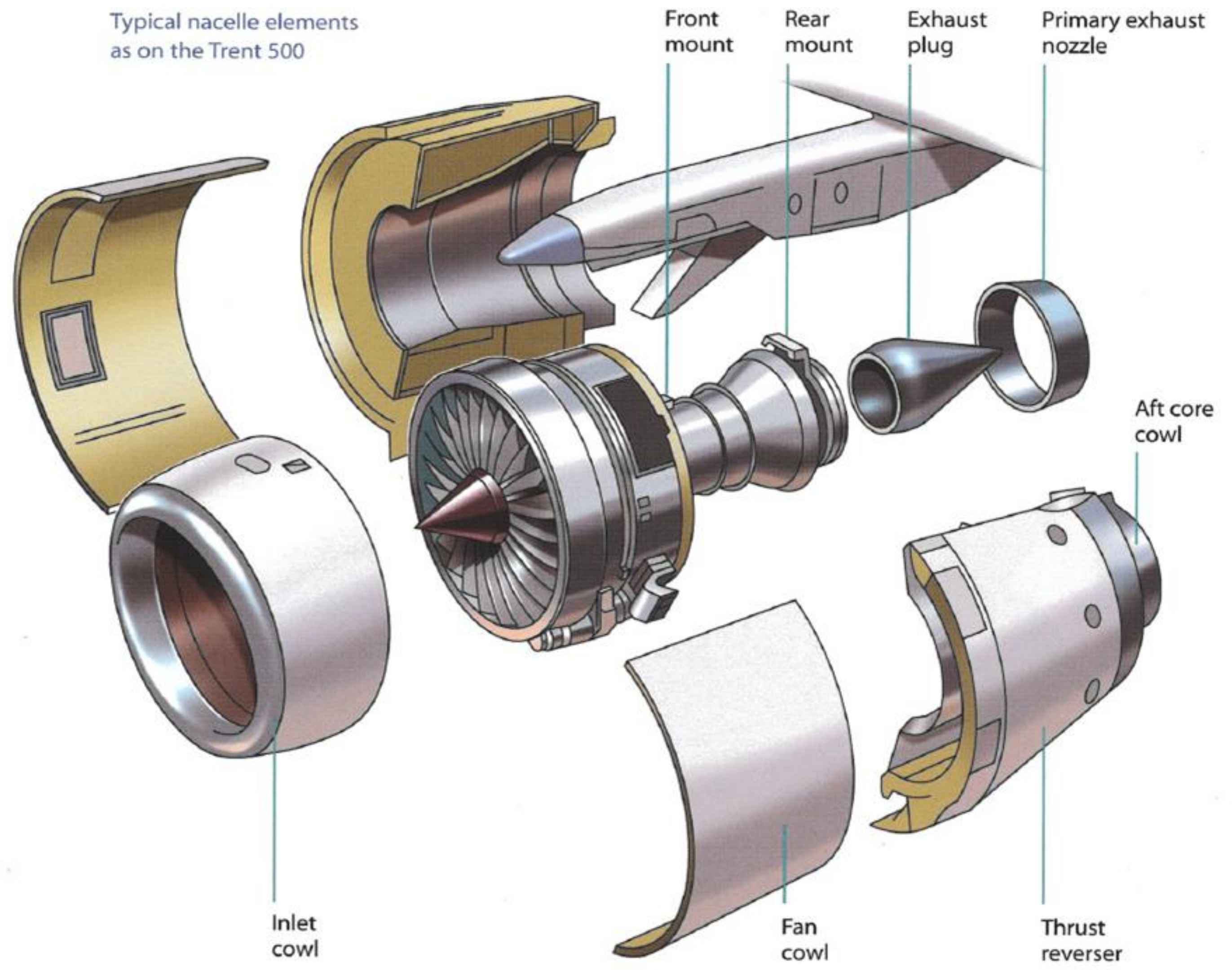
Typical nacelle of a commercial engine



Source: The Jet Engine, Rolls Royce PLC, 2006

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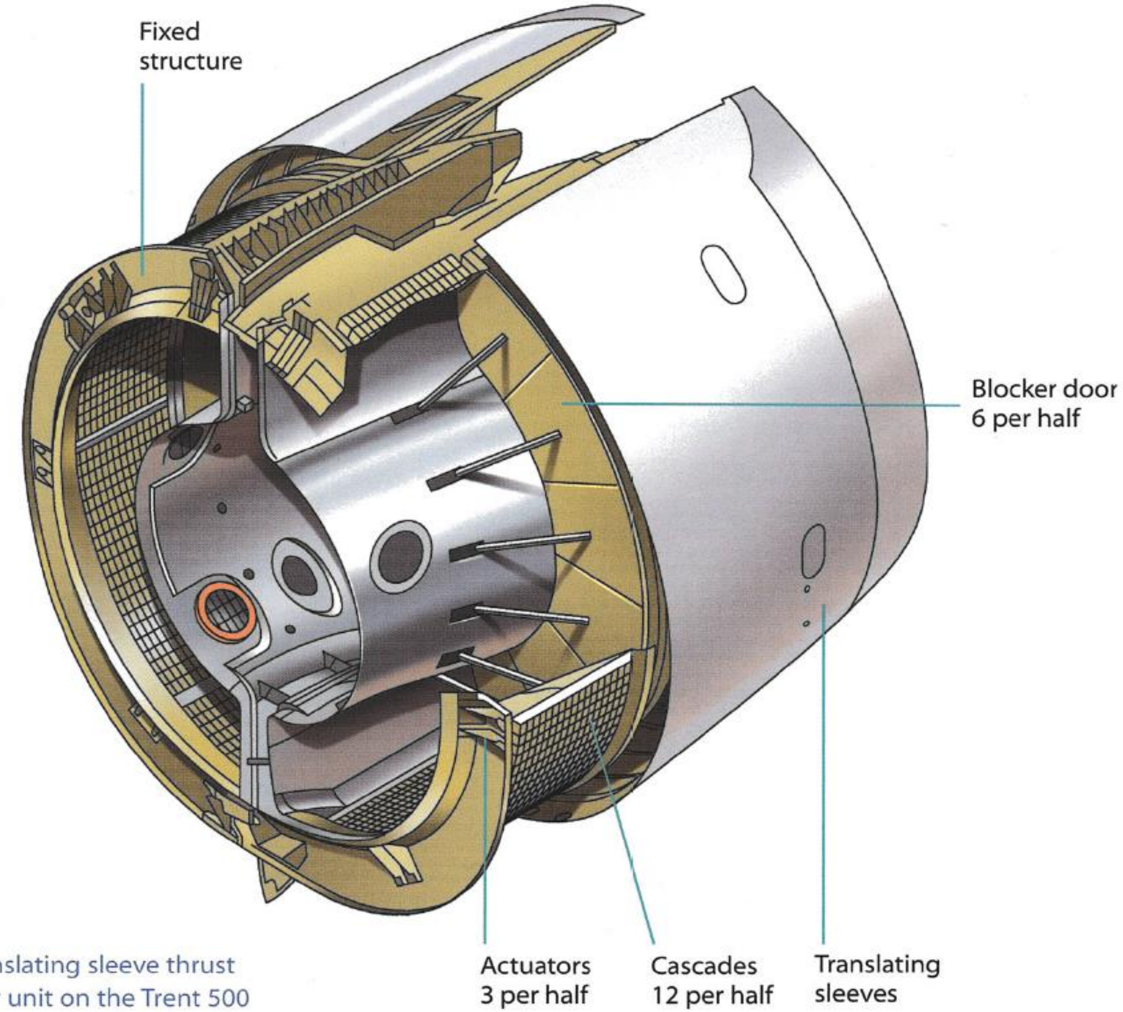
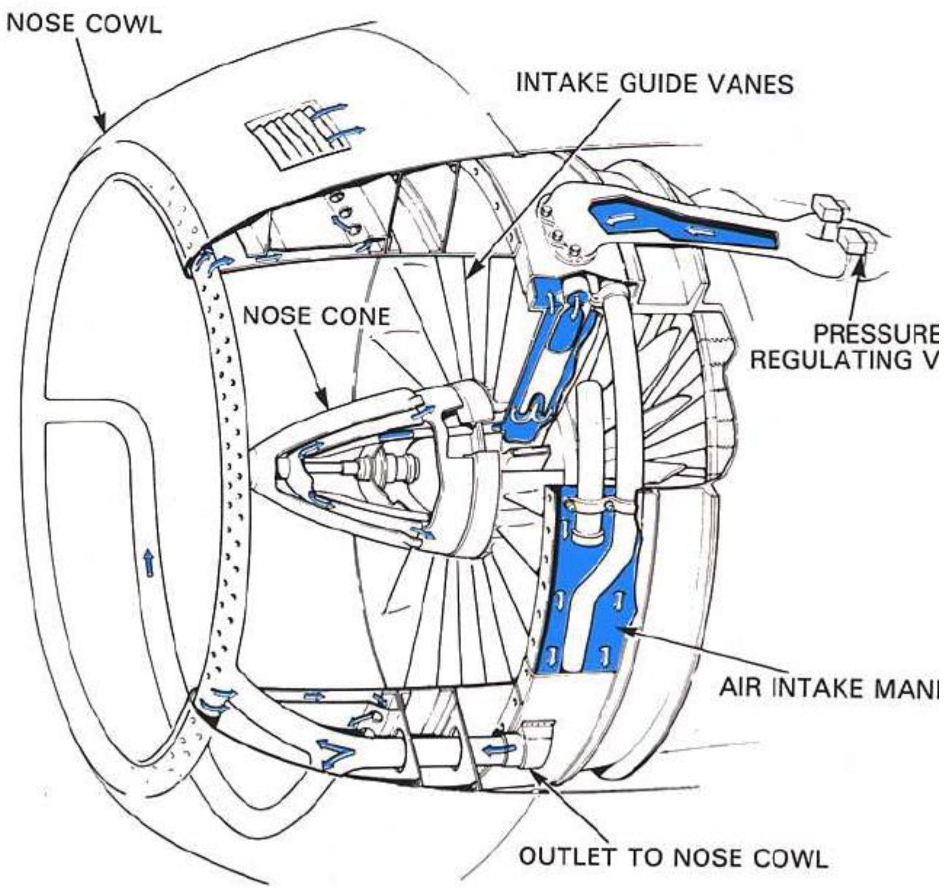
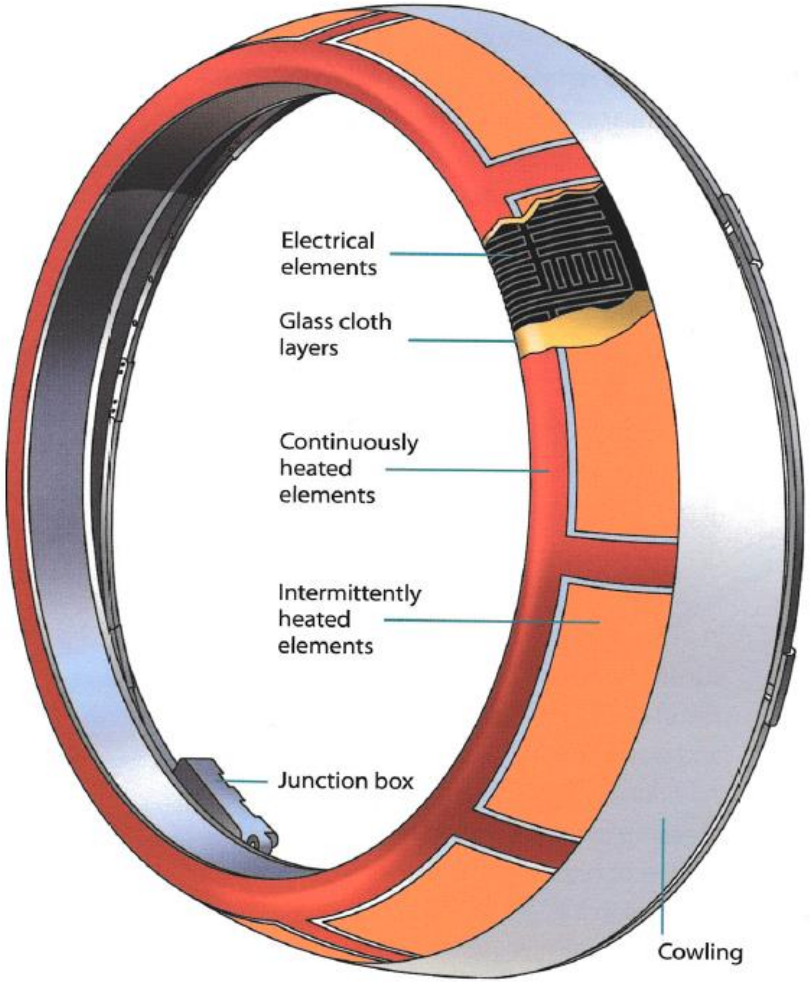
Thrust reversal



The translating sleeve thrust reverser unit on the Trent 500

Engine Intake





Source: “The Jet Engine”, Rolls Royce plc

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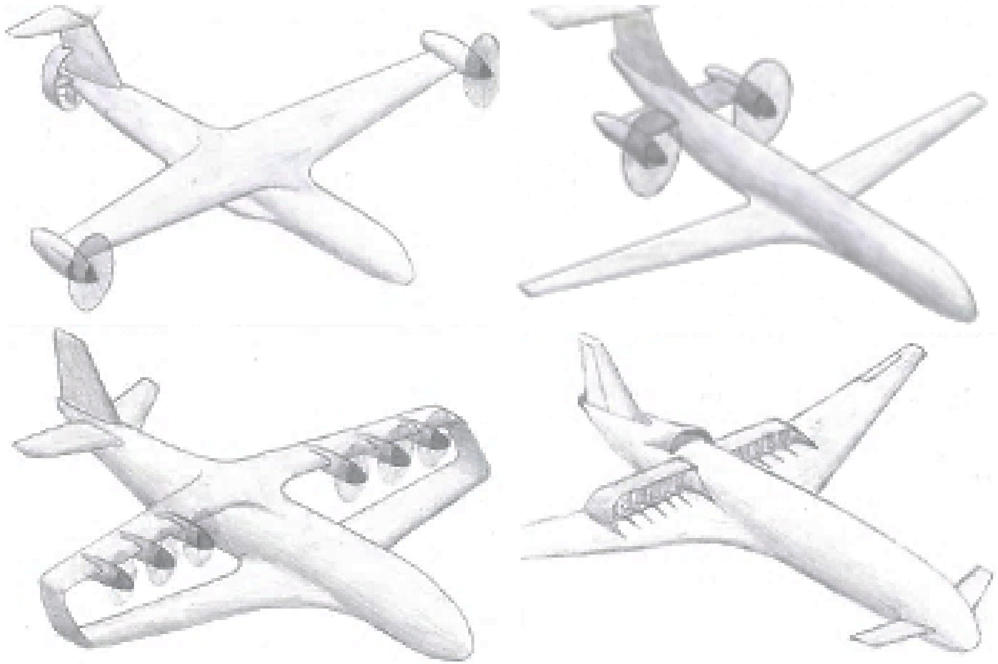
Thank You



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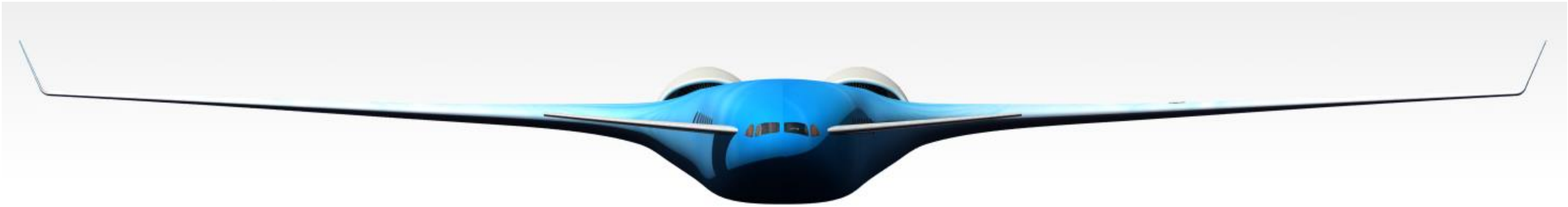

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